MILITARY SPECIFICATION

TEST SET SUBASSEMBLY: 10558483
(Card Reader)

1. SCOPE

This specification covers a test subassembly hereinafter called card reader, used in Test Set, Subassembly MX-8050/TPM-22, part of Radar Test Set AN/TP11-22.

2. APPLICABLE DOCUMENTS

The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military

MIL-F-13926	Fire Control Materiel: General Specification Governing the Manufacture and Inspection of
MIL-P-14232	Parts, Equipment and Tools for Army Material; Packaging and Packing of
MIL-I-45607	Inspection Equipment, Supply and Maintenance of
MIL-C-46357	Computer, Gun Direction; M18 Manufacturing Process and Inspection of

FSC 4931

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STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions
MIL-STD-810	Environmental Test Methods

DRAWINGS

U.S. Army, Frankford Arsenal

F10558483	Card Reader Assembly

PACKAGING DATA SHEET

10558483	Packaging of Card Reader Assembly

(Copies of specification, standards, drawings and packaging data sheets required by the suppliers-in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Fabrication. - The card reader shall be manufactured in accordance with Drawing F10558483 and drawings pertaining thereto.

3.2 General specification. - Where applicable, the card reader assembly shall meet the requirements of MIL-F-13926 and MIL-C-46357 as follows:

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3.2.1 MIL-F-13926

a. order of precedence
b. Dimensions and tolerances
c. Inorganic protective surface finishings
d. Part identification and marking
e. Workmanship
f. Soldering electrical connections

3.2.2 MIL-C-46357

a. Etched circuit assemblies
b. Wiring
c. Threaded fasteners
d. Marking
e. Finishes

3.3 Environmental service conditions.

3.3.1 Vibration. - The card reader assembly shall be mounted on the vibration table in normal service position. The card reader assembly shall meet the requirements of 3.4 and 3.5 after being subject to continuous vibration for two hours along each of three mutually perpendicular axes within the frequency range and amplitude as follows:

Frequency Range	Double Amplitude
5 to 15	0.06 inch
15 to 25	0.04 inch
25 to 55	0.02 inch

The rate of change of frequency at each frequency range shall not exceed one cycle-per-second per second (1 cps/see). Vibration tests shall be-conducted under ambient room conditions.

3.3.2 Shock. - The card reader assembly shall not suffer damage shall meet the requirements of 3.4 and 3.5 after being subjected to 18 impact shocks, consisting of three shocks in opposite directions along each of the three mutually perpendicular axes. The shock impulses shall have an acceleration of 15 gravity units (G's) and a pulse duration of 11 ± 1 milliseconds.

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3.3.3 Altitude.

3.3.3.1 Operating. - The card reader assembly shall meet the requirements of 3.4 and 3.5 during and after exposure to barometric pressure encountered at 10,000 feet above sea level (20.6 inches Hg).

3.3.3.2 Nonoperating. - The card reader assembly shall meet the requirements of 3.4 and 3.5 after exposure for not less than 1 hour to barometric pressure encountered at 40,000 feet above sea level (5.5 inches Hg).

3.3.4 Temperature.

3.3.4.1 Operating. - The card reader assembly shall meet the requirements of 3.4 and 3.5 when stabilized thermally at any temperature between +32°F and +131°F.

3.3.4.2 Nonoperating (storage). - The card reader assembly shall not suffer permanent damage when subjected to the temperature extremes of -80°F and +185°F, and shall meet the requirements of 3.4 and 3.5 when turned to room ambient.

3.3.5 Humidity-nonoperating. - The card reader assembly shall meet the requirements of 3.4 and 3.5 after exposure to an atmosphere with a relative humidity in excess of 95 percent for a period of 10 days.

3.5 Performance. - When supplied with +24 ± 0.5 V between connector terminals J3-A(RET) and J3-B(+), with a test card from Test Cards (11726471) for Test Set, Subassembly MX-8050/TPM-22 inserted in the card reader and the handle thrown to the righthand position, and with MATRIX SELECTOR 1 and MATRIX SELECTOR 2 both in position 12, switches shall be closed inside the card reader establishing circuits between the connector pins indicated in Table I. Each circuit so established shall be less than 0.3 ohms resistance, except where a specific resistance value is given in Table I.

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TABLE I - CIRCUITS ESTABLISHED BY CARD READER

TEST CARD	CIRCUITS ESTABLISHED FROM CONNECTOR PIN(S)	CIRCUITS ESTABLISHED TO CONNECTOR PIN(S)
ST 1	J5-A,B J4-S J4-17 J4-JJ	J1-A thru BN J4-R J4-W J4-KK
ST2	J5-D,E J4-V J4-JJ	J1-A thru BN J4-U J4-HH
ST 3	J2-A J4-Y	J1-A,B,C,D,E,F,H,J, J4-X
ST4	J2-D J4-Y J4-EE	J1-A,B,C,D,E,F,H,J J4-Z J4-FF
ST5	J2-H	J1-A,B,C,D,E,F,H,J
ST6	J2-L	J1-A,B,C,D,E,F,H,J
ST7	J2-P J3-K	J1-B,D,F,J J3-C,D
ST8	J2-T J3-K J4-EE	J1-B,D,F,J J3-E J4-DD
ST9	J2-W	J1-B,D,F,J
ST10	J2-Z	J1-B,D,F,J
ST11	J2-CC J5-K J4-C,F J3-K	J1-A,B,C,D J1-E,F,H,J J1-BT, J6-A J3-F

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TABLE I - CIRCUITS ESTABLISHED BY CARD READER

TEST CARD	CIRCUITS ESTABLISHED FROM CONNECTOR PIN(S)	CIRCUITS ESTABLISHED TO CONNECTOR PIN(S)
ST12	J5-N	J1-B,F,H,J
	J5-R	J1-B
	J7-A,D,H	J1-C
	J4-K	J1-BK,BL
	J4-M	J1-BH,BJ
	J4-J	J1-BM,BN
	J4-L	J1-BE,BF
	J4-E	J1-R
	J4-P	J1-AB
	J4-C,F	J1-A, Z,AA,AF,AN,AP, BT; J4-N
	J4-A	J1-D,AC,AR,AU
	J4-D	J1-X,Y
ST13	J6-D	J1-BP,BR; J6-A(24.95 ohms ±1%) J1-BT(24.95 ohms ±l%)
ST15	J6-L	J1-BP,BR
	J4-E	J1-BM,BN
ST16	J6-P	J1-BP,BR
	J4-P	J1-BM,BN
	J4-BB	J4-CC
ST17	J6-B	J1-BP,BR
	J4-H	J1-BM,BN
ST18	J5-H	J1-A,B,C,D,E,F,H
	J4-B	J1-J
	J4-E	J1-J(1100 ohms ±5%)
ST19	J1-BM	J1-BN
	J1-A	J1-B
	J6-A	J1-AF
ST20	J6-A	J1-BP(450 ohms ±2%)
	J6-A	J1-BR(450 ohms ±2%)
	J6-A	J1-BM(100 ohms ±l%)
	J6-A	J1-BN(100 ohms ±l%)

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TABLE I - CIRCUITS ESTABLISHED BY CARD READER

TEST CARD	CIRCUITS ESTABLISHED FROM CONNECTOR PIN(S)	CIRCUITS ESTABLISHED TO CONNECTOR PIN(S)
ST21	J6-A	J1-BM(1100 ohms ±5%)
ST22	J6-A	J1-BM(ll,100 ohms ±5%)
ST23	J3-K J4-S	J3-11 J4-T
ST27	J7-D	J1-A thru BN
ST28	J7-D	J1-A thru BN
ST29	J7-D	J1-A thru BN
ST30	J7-D	J1-A thru BN
ST32	J4-B3	J4-AA
ST34	J3-L J3-N	J3-M J3-P
ST35	J3-K	J3-J

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TABLE I - CIRCUITS ESTABLISHED BY CARD READER

TEST CARD	CIRCUITS ESTABLISHED FROM CONNECTOR PIN(S)	CIRCUITS ESTABLISHED TO CONNECTOR PIN(S)
ST14	J6-H	J1-BP,BR
	J6-A	J1-A
	J4-A	J1-BM,*BN**

* Resistance shall be as follows with MATRIX SELECTOR 1 in the following positions:

Position	Resistance
1	Open
2	100k ±5%
3	75k ±5%
4	33k ±5%
5	10k ±5%
6	7.5k ±5%
7	3.3k ±5%
8	1k ±1%
9	750k ±1%
10	332k ±1%
11	100k ±1%

**Resistance shall be as follows with MATRIX SELECTOR 2 in the following positions:

Position	Resistance
1	Open
2	100k ±5%
3	75k ±5%
4	33k ±5%
5	10k ±5%
6	7.5k ±5%
7	3.3k ±5%
8	1k ±1%
9	750k ±1%
10	332k ±1%
11	100k ±1%

Note: Table I does not exhaust the connections that can be established by the card reader, but it does include at least one connection through each switch in the assembly.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any other facilities suitable for the performance of the inspection requirements specified herein unless disapproved by the Government. The Government reserves the right-to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements .

4.1.1 General provisions . - The component and subassembly inspection requirements of MIL-F-13926 form a part of the Quality Assurance Provisions of this specification. Definitions of inspection terms shall be listed in MIL-STD-109.

4.2 First article (initial production) approval. - The requirement for first article approval and the responsibility (Government or contractor) for the first article testing shall be as specified in the contract. The sample for first article approval tests shall consist of one card reader assembly manufactured in the same manner, using the same materials, equipment, processes and procedures as used in regular production. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as used in regular production.

4.2.1 Government testing. - When the Government is responsible for conducting first article approval tests, the contractor, prior to submitting the sample to the Government, shall inspect the sample to ensure that it conforms to all the requirements of the contract and submit a record of this inspection with the sample, including certification of conformance for materials.

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4.2.2 Contractor testing. - When the contractor is responsible for conducting first article approval tests, the sample shall be inspected by the contractor for all the requirements of the contract. The sample and a record of this inspection, including certification for materials, shall be submitted to the Government for approval. The Government reserves the right to witness the contractor's inspection.

4.3 Inspection provisions.

4.3.1 Submission of product. - Unless otherwise specified by the Contracting Officer, inspection lot size, lot information, and presentation of lots shall be in accordance with "Submission of Product" provisions of MIL-STD-105.

4.3.2 Examination and tests. - Components and subassemblies shall be inspected in accordance with the inspection provisions contained in MIL-F-13926. Examinations and tests related to Section 3 herein shall be performed on a single defect (individual characteristic) basis in accordance with MIL-STD-105 and the sampling plans specified in Tables II and III herein. Examinations and test for Packaging, Packing and Marking shall be in accordance with MIL-P-14232 and Section 5 herein. The tabulated classification of defects shall constitute the minimum inspection to be performed by the supplier prior to Government acceptance or rejection by lot.

TABLE II - CLASSIFICATION OF DEFECTS

100% INSPECTION

CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
101. Electrical	3.4	4.6.2
102. Performance	3.5	4.6.3

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4.3.3 Acceptance and rejection. - Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in MIL-STD-105.

4.4 Special sampling. - One card reader assembly out of the first 50 produced and one out of each 100 produced thereafter shall be selected at random. The sample shall have met the requirements and tests in Table II and shall then meet the requirements and tests in Table III.

TABLE III - SPECIAL SAMPLING

	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
301.	Fabrication	3.1	Applicable dwg. visual
302.	General specifications	3.2	MIL-F-13926/ MIL-C-46357 visual
303.	Vibration	3.3.1	4.6.1.1
304.	Shock	3.3.2	4.6.1.2
305.	Altitude - operating	3.3.3.1	4.6.1.3
306,	Altitude - nonoperating	3.3.3.2	4.6.1.3
307.	Temperature	3.3.4	4.6.1.4
308.	Humidity - nonoperating	3.3.5	4.6.1.5

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4.4.1 Failure of sample. - Should any one item of a special sampling fail to meet the specified test requirements, acceptance of the product shall be suspended by Government until necessary corrections have been made by the contractor and a resubmitted sample has been approved (see 4.3.3).

4.5 Inspection equipment. - Except as otherwise provided for by the contract, the contractor shall supply and maintain inspection equipment in accordance with the applicable requirements of MIL-I-45607.

4.5.1 Government-furnished inspection equipment. - Where the contract Provides for Government-furnished test equipment, supply and maintenance of test equipment shall be in accordance with the applicable requirements of MIL-I-45607.

4.5.2 Contractor-furnished inspection equipment.

4.5.2.1 Contractor design. - The contractor shall design and supply inspection equipment compatible with "Test Methods and Procedures" specified in 4.6 of this specification and with the "Component Inspection" procedures specified in "Examination" and "Test Facilities" requirements of MIL-F-13926. Since tolerance of test equipment is considered normally to be within 10 percent of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus , concept, construction, materials, dimensions, and tolerances used in design of test equipment shall be so selected and controlled as to ensure that the test equipment will indicate reliable acceptability of a product that does not exceed 90 percent of the prescribed tolerance limit and permit positive rejection when nonconforming. Construction shall be such as to facilitate routine calibration of the test equipment.

4.5.2.2 Test equipment. - In conjunction with 4.5.2.1, the following test equipment shall be used in the performance of the applicable test as specified in 4.6:

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TEST EQUIPMENT LIST

Item		Description
1.	Power supply	24.0 ± 1 Vdc, capacity 2A.
2.	Kelvin bridge	Capable of measuring resistances from 0.01 ohms to 1.0 ohm, accuracy 1% of reading plus 1% of full scale of scale being used,
3.	Ohmmeter	Capable of measuring resistances from 1.0 ohm to 1 megohm; accuracy 0.5% of reading plus 0.5% of full scale, for values up to 1 kilohm; accuracy 2% of reading plus 1% of full scale for values above 1 kilohm.
4.	Test harness	Per Figure 1.

PARTS LIST FOR TEST HARNESS, Item 4

Designation	Part	Description
P1	Connector	Elco 00-8016-090-217-703 or equivalent
P2	Connector	Elco 00-8016-038-217-723 or equivalent
P3	Connector	Elco 00-8016-020-217-709 or equivalent
P4	Connector	Same as P2
P5	Connector	Same as P3
P6	Connector	Same as P3
P7	Connector	Same as P3
Terminal strips, wire, per Figure 1.		

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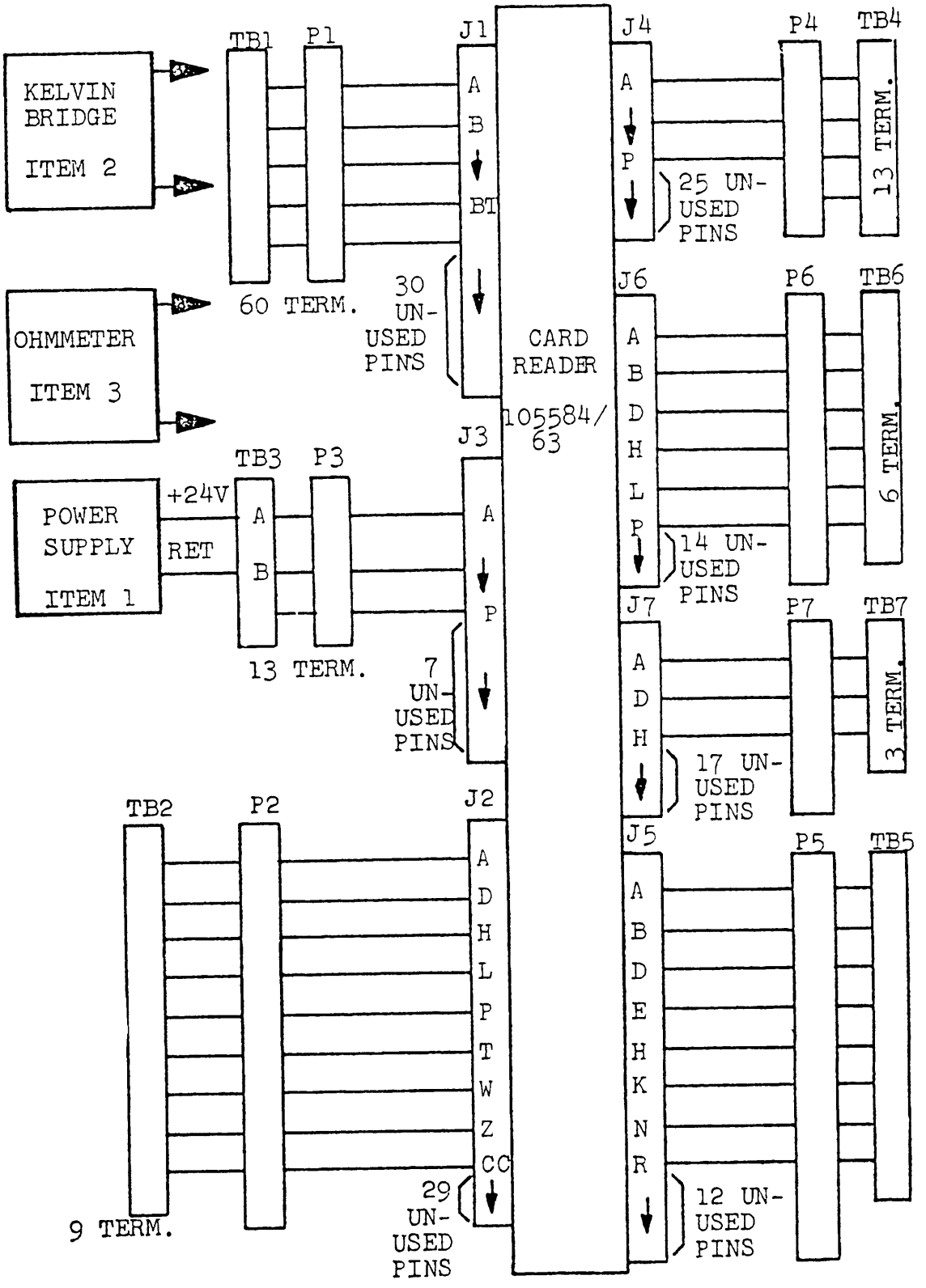


FIGURE 1 CONNECTIONS TO CARD READER

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4.6 Test methods and procedures.

4.6.1 Environmental. - Environmental test shall be conducted in accordance with the general requirements and test methods of MIL-STD-81O except as specified herein. Testing equipment used in environmental tests shall be in accordance with “Test Facilities and Apparatus” requirements of MIL-STD-81O.

4.6.1.1 Vibration. - The test shall be performed as specified in Method 514, Procedure IX, Part I, of lIIIL-STD-81O except that Failure Criteria of Method 514 of MIL-STD-81O do not apply. After vibration, the card reader assembly shall be tested as given in the sub-paragraphs of 4.6.2 and 4.6.3 for compliance with 3.3.1.

4.6.1.2 Shock. - This test shall be performed as specified in Procedure I of Method 516 of MIL-STD-81O except that shock pulses shall be 11 ± 1 ms and 15 g’s. The G value of each impulse shall be within ±10 percent when measured with a 0.2 to 250 Hz filter and the maximum G shall occur at approximately 5-1/2 milliseconds. After the shock impulses, the card reader assembly shall be tested as specified in 4.6.2 and 4.6.3 for compliance with 3.3.2.

4.6.1.3 Altitude - non-operating and operating. - This test shall be performed as specified in Procedure I, Method 500 of MIL-STD-81O except that in Step 2 the altitude shall be limited to 40,000 ft. (5.5 in Hg). Operation of Step 3 shall be the performance tests of 4.6.3.

4.6.1.4 Temperature.

4.6.1.4.1 High (operating and non-operating). - This test shall be performed as specified in Procedure I, Method 501, of MIL-STD-81O. The temperature of Step 2 shall be +185°F. The temperature of Step 4 shall be +130°F. In Step 5 the card reader assembly shall be tested in accordance with 4.6.2 and 4.6.3.

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4.6.1.4.2 Low (operating and non-operating) . - This test shall be performed as specified in Procedure I, Method 502 of MIL-STD-810. The temperature of Step 2 shall be -85°F maintained for 24 hours. The temperature of Step 4 shall be +32°F. In Steps 5 and 7, the card reader assembly shall be tested in accordance with 4.6.2 and 4.6.3.

4.6.1.5 Humidity-nonoperating. - This test shall be performed in accordance with Procedure 1, Method 507 of MIL-STD-810 except that in Step 4, the relative humidity shall be maintained at 95%.

4.6.2 Electrical. - Continuity shall be inspected by standard measuring equipment (SME) and visual examination for compliance with 3.4-.

4.6.3 Performance. - Adjust power supply, item 1, to deliver +24 ±0.5 V and connect to test harness, item 4, as shown in Figure 1. Insert Self Test Card #l (ST1) in card reader slot and throw handle to righthand position. use Kelvin bridge, item 2, to measure resistance between the appropriate terminals specified in Table I, via the terminal strips of test harness, item 4. Each resistance measurement shall be less than 0.3 ohms. Repeat this procedure for each Self Test Card as indicated in Table I. Where a specific resistance value is given in Table I, use ohmmeter, item 3, instead of Kelvin bridge, and measure value given in Table I.

5. PREPARATION FOR DELIVERY

5.1 Packaging. - Packaging, packing, and marking shall be in accordance with Packaging Data Sheet 10558483. The level of protection shall be as specified in the contract.

6. NOTES

6.1 Intended use. - The card reader contains a matrix of switches which is used to establish any desired connection between two or more terminals of the matrix, or two or more separate connections. These connections are established when an appropriately punched card is inserted in the card reader slot and the actuating handle on the card reader is thrown to the proper position. The card reader is used with other equipment to apply desired voltages, signals, and loads to selected terminals of printed circuit cards in the course of fault detection and isolation.

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6.2 ordering data. - Procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) Applicable packaging data sheet (see 5.1).

(c) Selection of applicable level of preservation packaging and packing.

(d) Level of protection required.

(e) Applicable stock number.

(f) Block of serial numbers to identify each card reader assembly.

Custodian	Preparing activity
Army - MU	Army - MU
	Project No. 4931-A030

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE | CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

YES [ ] NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)* | DATE

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